

From the Boundary of the Firm to the Boundary of the Agent

Coase–Information Theory for AI-Mediated Organizations

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Abstract

Coase explained the firm as an alternative to market coordination when using the price mechanism is costly. This paper extends that design problem from the boundary of the firm to the boundary of the agent: a bounded sensing–representation–decision–action loop that may comprise humans, teams, services, vendors, or software agents. It develops a normative framework for choosing such boundaries under a specified objective, decomposes coordination costs into operational terms, and defines organizational agility as decision value per unit time. Established information-theoretic tools yield an analytical benchmark: for a memoryless scalar Gaussian estimation task with quadratic loss and asymptotic one-way coding, the excess loss of splitting is the zero-communication decision deficit multiplied by $2^{-2\kappa}$. Combining that benchmark with coordination savings gives a finite economic switching threshold, even when preserving all decision value requires unbounded communication. Under balanced units, quadratic internal cost, and sparse constant-cost interfaces, a separate continuous relaxation gives the familiar square-root granularity rule. Agents can favor either finer decomposition or greater integration, depending on their relative effects on internal costs, interface costs, and decision loss. The contribution is the integration of these results into a recursive boundary-design framework, not a new general coding theorem or universal scaling law. A stylized companion and a controlled-replay research design distinguish analytical benchmarks from operational proxies and empirical evidence. Software agents are boundary-shifting infrastructure because programmable observation, communication, and action change where shared context is worth its cost.

Keywords: multi-agent systems; organizational design; transaction-cost economics; software agents; information processing; organizational agility

1 Introduction: The Boundary Problem After Software Agents

Coase framed the firm as a response to the costs of using the price mechanism: firms exist because contracting, searching, negotiating, monitoring, and enforcing every transaction through the market can be more costly than directing work internally [1]. That framing remains essential, but it is no longer sufficient for organizations whose coordination is mediated by software systems, APIs, observability pipelines, ticket queues, chat systems, workflow engines, and autonomous software agents.

These tools do not merely lower generic transaction costs. They change what can be observed, how information is compressed, which interfaces are machine readable, how quickly actions can be taken, and where governance must be applied. The resulting design question is:

Given a changing environment, what boundary architecture best converts distributed, noisy information into coordinated action under latency, cost, and governance constraints?

This question applies at multiple levels. It determines whether work should be inside a firm or outside it. It also determines whether a team should centralize into an incident room, split into modular services, coordinate with a vendor, or use software agents to mediate between otherwise separate actors. A firm is one boundary choice. A department, workflow, API contract, vendor interface, software service, and autonomous agent are others.

The central claim is:

Firms, teams, vendors, workflows, and software agents are boundary choices over information-processing units. Boundaries form when shared representation and control loops reduce surprise, delay, error, or misalignment more than they add coordination and governance cost.

The contribution is not the broad claim that firms process information. That intuition has deep roots in Hayek, Simon, Galbraith, and distributed decision theory [6, 7, 15, 11]. Formal organizational economics has derived structures from communication, processing, and incentive costs [18, 19, 21, 22]. This paper integrates those perspectives into a recursive design language for human–software organizations. Programmable interfaces can change both the cost of pooling information and the cost of communicating across a boundary; neither direction is assumed to dominate.

Three analytical steps organize the argument. An operational capacity threshold defines preservation of a specified decision value (Proposition 1). A Gaussian–quadratic example applies classical source coding with decoder side information to price excess decision loss (Theorem 1). Combining that loss with coordination savings gives the economic switching threshold (Corollary 2): information loss can be worth accepting. A separate balanced-unit cost model gives a square-root granularity rule (Theorem 2), with explicitly conditional comparative statics. The contribution is their integration and organizational interpretation, not novelty of the underlying coding or optimization machinery.

The framework is normative: it compares architectures under a specified objective, feasible policy set, and governance constraints. It does not derive bargaining equilibria, endogenous property rights, or the distribution of surplus. Misalignment and monitoring are costs to be specified, not a substitute for an incentive-compatibility model. Observed organizational boundaries need not coincide with the optima described here.

This paper does not claim that firms disappear, that decentralization is always superior, or that software agents universally improve coordination. The claim is conditional: software agents alter the relative costs of observation, compression, routing, interpretation, execution, monitoring, and misalignment. Depending on task structure, protocol quality, observability, and governance, they may favor coalescence, splitting, or hybrid coordination.

2 Related Intellectual Lineage

The paper sits at the intersection of four literatures.

First, transaction-cost economics explains why activities are sometimes organized inside firms rather than across markets [1, 16]. Search, contracting, monitoring, opportunism, and asset specificity all shape the efficient boundary of the firm.

Second, Hayek’s account of dispersed knowledge and Simon’s account of bounded rationality treat organizations as mechanisms for coping with limited, distributed information [6, 7, 9, 10]. Marschak and Radner’s economic theory of teams formalizes organizations as distributed decision systems [11]. Galbraith’s information-processing view of organizations links uncertainty, task interdependence, and structural design [15]. Coordination theory and classic organization theory similarly treat coordination as the management of dependencies among activities [14, 17]. This tradition has a precise formal counterpart. Radner models hierarchies as parallel information processors trading processing delay against managerial cost [18], and Van Zandt extends the analysis to real-time processing by boundedly rational agents [20]. Bolton and Dewatripont derive efficient communication networks when absorbing messages is costly [19]; Garicano derives knowledge hierarchies that economize on acquiring and transmitting knowledge [21]; and Alonso, Dessein, and Matouschek characterize when coordination requires centralization under strategic communication [22]. Arrow’s information-cost account of organization [12], Simon’s near-decomposability [8], and Brooks’s all-pairs communication burden [13] anticipate the cost structure used in Section 7. These models provide the foundation for treating communication and processing technologies as organizational design parameters. The present paper emphasizes programmable interfaces and recursively composed human–software decision units, without claiming that technological comparative statics are absent from the earlier literature.

Third, information theory provides a vocabulary for representation, compression, fidelity, and distortion [2]. That vocabulary is useful, but not sufficient by itself, because organizational action is valuable only through the loss or utility it produces. Blackwell’s comparison of experiments provides the relevant informativeness language: one information structure is more useful when every decision-relevant statistic available under the old structure can be reconstructed under the new one, and possibly more [3].

Fourth, modern software systems introduce programmable protocols, observability streams, APIs, queues, schemas, automated routing, and increasingly autonomous software agents. These make some cross-boundary coordination machine-readable and partially executable. This connects the theory to modularity, information hiding, architectural design, and socio-technical structure in software systems [23, 24, 25, 26], as well as to recent work on tool-using and interactive AI agents [27, 28, 29, 30, 31, 32].

The proposed synthesis combines these threads into a single recursive boundary-design problem. The unit of analysis is not only the legal firm. It is any bounded sensing-representation-decision-action loop.

3 Agent-Boundary Model

Let the external or internal environment generate a latent state

$$X_t \in \mathcal{X}$$

at time t . The state may include customer demand, an incident state, a supply shock, a workflow backlog, a vendor outage, or any other decision-relevant facts.

An organization is a directed graph

$$G = (V, E),$$

where nodes $i \in V$ are humans, teams, services, vendors, software agents, or composite groups. Edges $(i, j) \in E$ are communication or coordination channels.

The word *agent* is used recursively. A node may be a primitive actor, such as a person or software service, or a composite actor, such as a team, firm, platform, vendor network, or incident-response group. Let V_0 be the set of primitive actors and let

$$\mathcal{B} = \{B_1, \dots, B_m\}$$

be a partition of V_0 into bounded agents. Each block B_ℓ is an agent when its members share enough representation, policy, governance, and action interfaces to be treated as one decision-making unit at the relevant level of analysis.

More formally, let $G_0 = (V_0, E_0)$ denote the graph of primitive actors and communication channels. A boundary partition $\mathcal{B} = \{B_1, \dots, B_m\}$ induces a quotient graph $G_{\mathcal{B}}$, whose nodes are bounded agents and whose edges are cross-boundary coordination channels. For readability, the paper writes $G = (V, E)$ for the graph at the level of analysis under discussion.

Each node receives a local observation

$$Y_t^i \sim P_i(Y \mid X_t)$$

and may send messages

$$M_t^{ij} \in \mathcal{M}_{ij}$$

to neighboring nodes. Messages may be unstructured human communication, structured tickets, API calls, database writes, workflow events, or agent-generated summaries.

Each node maintains a compressed local representation

$$Z_t^i = f_i(Y_{\leq t}^i, M_{\leq t}^{*i}),$$

where $M_{\leq t}^{*i}$ denotes messages received by node i . Node i chooses an action according to a local policy

$$a_t^i \sim \pi_i(Z_t^i).$$

The organization produces a joint action

$$a_t = g((a_t^i)_{i \in V})$$

and receives utility $U(X_t, a_t)$, or equivalently incurs loss $L(X_t, a_t) = -U(X_t, a_t)$. Unless otherwise stated, L denotes direct task or decision loss from the chosen action; coordination, delay, monitoring, interpretation, and governance burdens enter through separate cost terms so that they are not counted twice.

All cumulative quantities below refer to a common finite evaluation horizon and case distribution; infinite-horizon applications require an explicit discount or average-cost convention. The general organizational design problem is

$$\max_{\mathcal{B}, G, \Pi, F} \mathbb{E} \left[\sum_t U(X_t, a_t) \right] - C_{\text{comm}} - C_{\text{comp}} - C_{\text{interp}} - C_{\text{delay}} - C_{\text{error}} - C_{\text{misalign}} - C_{\text{monitor}},$$

where $\Pi = \{\pi_i\}$ is the set of local policies and $F = \{f_i\}$ is the set of compression or representation maps. The same problem can be written in loss form by replacing $U(X_t, a_t)$ with $-L(X_t, a_t)$, turning the maximization of utility minus costs into a minimization of loss plus costs. The scalarization J introduced below generalizes this by attaching explicit weights to each cost term and additionally pricing surprise.

4 Cost Decomposition and Boundary Performance Vector

The Coase–Information move is to treat transaction and coordination costs as partly information-processing quantities. Table 1 gives the minimal decomposition used in the rest of the paper.

Table 1: Information-processing decomposition of coordination cost

Term	Interpretation
C_{comm}	Meetings, messages, handoffs, routing, API calls, protocol overhead.
C_{comp}	Human or machine effort to analyze, summarize, decide, or execute.
C_{interp}	Translation and interpretation across boundaries: schema mismatch, jargon, missing context, tool-output interpretation, vendor/customer translation, and semantic ambiguity.
C_{delay}	Incremental process loss from acting after the state has changed or the opportunity has decayed.
C_{error}	Incremental process loss from stale, incomplete, distorted, or misrouted information.
C_{misalign}	Incremental process loss from local actions that do not compose into good global action.
C_{monitor}	Measurable governance burden: audit, review, compliance, approval, and oversight steps.

This table is intentionally operational. The quantities can be estimated from workflow traces, incident response logs, ticket systems, Slack histories, API calls, approval queues, and agent tool traces. It also prevents the paper from treating “coordination cost” as a black box. A boundary change can reduce one cost term while increasing another. A software agent that removes routing delay but increases monitoring burden is not automatically beneficial; it changes the cost vector. In this paper, monitoring cost is the measurable component of governance cost: approvals, audits, reviews, escalation requirements, and oversight work.

Formally, each boundary architecture induces a performance vector

$$\Phi(\mathcal{B}, G, F, \Pi) = (\mathbb{E} \sum_t S_t, \mathbb{E} \sum_t L(X_t, a_t), C_{\text{comm}}, C_{\text{comp}}, C_{\text{interp}}, C_{\text{delay}}, C_{\text{error}}, C_{\text{misalign}}, C_{\text{monitor}}).$$

Here $\mathbb{E} \sum_t S_t$ is a predictive score evaluated on a fixed set of targets, with the same scoring opportunities and weights for every architecture (Section 6). It is not a sum over a changing number of agents. The term $\mathbb{E} \sum_t L(X_t, a_t)$ is cumulative action loss.

The scalar objective used by a particular organization is a weighted or constrained evaluation of this vector. One simple scalarization is

$$\begin{aligned} J(\mathcal{B}, G, F, \Pi) = & \mathbb{E} \sum_t L(X_t, a_t) + \beta_S \mathbb{E} \sum_t S_t \\ & + \lambda_{\text{comm}} C_{\text{comm}} + \lambda_{\text{comp}} C_{\text{comp}} + \lambda_{\text{interp}} C_{\text{interp}} + \lambda_{\text{delay}} C_{\text{delay}} \\ & + \lambda_{\text{error}} C_{\text{error}} + \lambda_{\text{misalign}} C_{\text{misalign}} + \lambda_{\text{monitor}} C_{\text{monitor}}. \end{aligned}$$

Weights are nonnegative and costs use a common declared accounting convention. By default $\beta_S = 0$: predictive surprise is a diagnostic unless an independent economic burden, not already in loss or other costs, justifies pricing it. Then an objective-efficient architecture satisfies

$$(\mathcal{B}^*, G^*, F^*, \Pi^*) \in \arg \min_{\mathcal{B}, G, F, \Pi} J(\mathcal{B}, G, F, \Pi).$$

This makes the tradeoff explicit: boundary design is multi-objective, but actual organizational choice requires weights, constraints, or governance priorities.

5 Pareto and Objective-Efficient Boundaries

Definition 1 (Agent boundary). *An agent boundary is a partition of actors, observations, representations, policies, tools, permissions, and actions into an internal control loop plus external interfaces. Each boundary induces a performance vector consisting of expected surprise, action loss, communication cost, computation cost, interpretation cost, delay cost, error cost, misalignment cost, and monitoring cost. A boundary is Pareto-efficient when no feasible architecture weakly improves every component of this vector and strictly improves at least one. A boundary is objective-efficient for a given organization when no feasible architecture lowers the organization’s total expected objective after applying its weights, constraints, or governance requirements over these quantities.*

Pareto efficiency describes the frontier of feasible boundary architectures. With strictly positive weights on all components, global objective minimizers are Pareto-efficient on the same feasible set. With zero weights, a minimizer can be dominated on ignored components; use Pareto-efficient tie-breaking when that distinction matters. Efficiency against only local merge/split moves is local optimality, not global Pareto efficiency. Different organizations may choose different efficient boundaries because they assign different weights to speed, accuracy, compliance, safety, morale, resilience, and strategic learning. In an incident context, a regulated financial platform may choose more monitoring and slower action than a consumer application even when both face the same technical incident, because their governance weights differ.

6 Coalescence, Splitting, and Protocol-Mediated Coordination

The boundary of an agent determines which observations, representations, and actions are internal to a shared control loop and which must cross an interface. Internal coordination can preserve richer context and reduce surprise, but it also creates communication, governance, computation, and planning costs. External coordination can reduce internal burden and preserve modularity, but it can increase interpretation, monitoring, contracting, and delay costs.

Let Z_t^B denote the internal representation held by a bounded agent B , and let a_t^B denote the action induced by that boundary’s representation and policy.

A local diagnostic of surprise is

$$S_t^B = -\log P_B(X_t \mid Z_t^B).$$

For continuous targets this denotes a density-based log score relative to a fixed reference measure. Local scores help diagnose prediction failures, but summing them over $B \in \mathcal{B}$ is not an architecture-neutral objective: merging two identical predictors would remove a term without improving any prediction. Changing a continuous state’s units would also add a constant per agent, making the comparison depend on the number of blocks.

Instead fix a set $\mathcal{D}$ of decision or prediction targets, fixed weights $w_j \geq 0$, and fixed evaluation times, independently of the architecture. Let $X_t^{(j)}$ be target j , and let $\hat{P}_{j,t}^{\mathcal{B}}$ be the predictive distribution delivered at that fixed evaluation interface by architecture $\mathcal{B}$. Define

$$S_t(\mathcal{B}) = \sum_{j \in \mathcal{D}} w_j \left[-\log \frac{d\hat{P}_{j,t}^{\mathcal{B}}}{d\nu_j}(X_t^{(j)}) \right],$$

using common reference measures ν_j , a common log base, and finite expected scores. For discrete targets ν_j is counting measure. A common change of units now shifts scores equally across architectures rather than per block. Duplicate attention or prediction-maintenance work, when genuinely costly, belongs in measured computation or monitoring costs. A boundary-design problem can therefore be written

$$\min_{\mathcal{B}, G, F, \Pi} \mathbb{E} \sum_t L(X_t, a_t) + \beta_S \mathbb{E} \sum_t S_t(\mathcal{B}) + C(\mathcal{B}, G, F, \Pi),$$

with $\beta_S = 0$ unless separately justified.

Surprise and action loss play different roles. In some applications, surprise is an intermediate diagnostic quantity: unexpected states predict future action loss, rework, escalation, or delay. In those cases, S_t should be treated as a proxy or leading indicator rather than double-counted with L . In other applications, surprise is itself costly because it consumes attention, triggers escalation, creates risk, or forces expensive re-planning. The empirical specification should decide whether surprise enters as an independent cost term or as a measurement proxy for future loss. The same convention applies to delay, error, and misalignment costs: they should be interpreted as incremental process losses not already included in $L(X_t, a_t)$, or else absorbed into L and omitted as separate terms.

Two agents should coalesce when the merged boundary lowers expected surprise or action loss more than it raises internal coordination and governance cost:

$$\mathbb{E}[L_{\text{merged}}] + \beta_S \mathbb{E}[S_{\text{merged}}] + C_{\text{merged}} < \mathbb{E}[L_{\text{separate}}] + \beta_S \mathbb{E}[S_{\text{separate}}] + C_{\text{separate}}.$$

They should split when the inequality reverses.

In incident response, an incident room coalesces SRE, support, product, and infrastructure into one temporary agent so that partial diagnoses become a shared representation instead of crossing weak interfaces.

Let k index a coordination form:

$$k \in \{\text{internal, market, hybrid}\}.$$

In the general performance vector, search and contracting burdens are folded into communication, interpretation, monitoring, and delay costs as appropriate. When comparing coordination forms, it is useful to write them separately:

$$C_k = C_{\text{search},k} + C_{\text{contract},k} + C_{\text{comp},k} + C_{\text{comm},k} + C_{\text{interp},k} + C_{\text{monitor},k} + C_{\text{delay},k} + C_{\text{error},k} + C_{\text{misalign},k}.$$

The efficient boundary is selected by

$$k^* = \arg \min_k \{\mathbb{E}[L_k] + C_k\}.$$

Classical hierarchy reduces some contracting and search costs but can increase internal communication delay and bureaucratic compression. Market coordination can reduce internal burden but increase search, contracting, monitoring, and interpretation costs. Agent-mediated hybrid coordination can reduce cross-boundary interpretation and delay if protocols and governance are strong enough.

The claim is not that agents make firms smaller or markets always better. The claim is that software agents and machine-readable protocols can alter enough cost terms to create a new efficient region between classic hierarchy and classic market contracting.

7 Information Preservation and Economic Boundaries

Information preservation and economic efficiency are different questions. An interface can be lossy yet worth using, or lossless yet prohibitively expensive. This section first defines preservation, then prices a deficit in an analytically tractable case, and finally compares that deficit with the coordination savings from drawing a boundary.

7.1 An operational preservation threshold

Fix a joint distribution of a scalar or vector state X and pooled local observations (Y^A, Y^B) , a loss L , and an admissible policy class. All architectures are evaluated on that same problem. In the coalesced benchmark, the deciding side holds both observations. In the split architecture it holds Y^A and a one-way message from an encoder that observes Y^B . Define

$$\mathcal{L}_{\text{coal}} = \inf_{\pi} \mathbb{E}[L(X, \pi(Y^A, Y^B))], \quad \mathcal{L}_{\text{split}}(r) = \inf_{M \in \mathcal{P}(r), \pi} \mathbb{E}[L(X, \pi(Y^A, M))].$$

Here $\mathcal{P}(r)$ is a nested family of protocols of rate at most r . For a one-shot model this can mean a fixed-length message budget. For a memoryless block model it means an asymptotic description rate per sample, with average loss per sample. These are distinct operational problems: fix one convention before comparing r with interface capacity κ . The deciding side's observation is decoder side information. General remote decision problems need their own distortion/rate formulation; they are not all ordinary source-reconstruction problems [4].

The coalesced policy class is assumed to include emulation of every split protocol, including its randomization. Thus $\mathcal{L}_{\text{split}}(r) \geq \mathcal{L}_{\text{coal}}$.

Definition 2 (Decision-relevant interface rate). *The preservation threshold and its tolerance version are*

$$R^* = \inf\{r \geq 0 : \mathcal{L}_{\text{split}}(r) = \mathcal{L}_{\text{coal}}\}, \quad R_{\varepsilon}^* = \inf\{r \geq 0 : \mathcal{L}_{\text{split}}(r) \leq \mathcal{L}_{\text{coal}} + \varepsilon\},$$

with the infimum of an empty set equal to $+\infty$.

Proposition 1 (Capacity dichotomy). *Assume nested protocol classes and attainment of both the required loss infima and the lower endpoint defining R^* when finite. If $\kappa \geq R^*$, some feasible protocol attains coalesced risk. If $\kappa < R^*$, the optimal excess loss*

$$\Delta(\kappa) = \mathcal{L}_{\text{split}}(\kappa) - \mathcal{L}_{\text{coal}}$$

is positive and non-increasing with κ .

Proof. Nested feasible sets make optimal split risk non-increasing. Endpoint attainment gives the first claim. A zero gap below R^* would put that rate in the defining set, a contradiction. $\square$

Without attainment, use explicit rate/loss slack and distinguish an infimum from an achieved protocol. The proposition follows from the operational definition; it fixes terminology, not a novel coding theorem. It does not imply a kink, a discontinuity, or a measurable elbow in the loss curve. An ε -threshold may simply be a smooth tolerance crossing.

A cut is *information-clean* relative to a specified decision problem when it preserves coalesced risk. Conditional mutual information can help characterize dependence, but is not a universal achievable communication rate for an arbitrary decision loss. Neither low mutual information nor low R^* alone identifies an economically optimal partition.

7.2 A Gaussian–quadratic benchmark

Theorem 1 (Exponential interface-deficit benchmark). *Let independent decision samples be drawn from a fixed jointly Gaussian (X, Y^A, Y^B) , with scalar X , finite covariance, and quadratic loss $L(x, a) = (x - a)^2$. The encoder observes $(Y^B)^n$ and sends at most $n\kappa + o(n)$ bits; the decoder observes $(Y^A)^n$. With unrestricted measurable encoding/decoding and asymptotic average loss as $n \rightarrow \infty$,*

$$\Delta(\kappa) = \underbrace{[\text{Var}(X | Y^A) - \text{Var}(X | Y^A, Y^B)]}_{\Delta(0)} 2^{-2\kappa}.$$

If $\Delta(0) > 0$, then $R^* = \infty$ and, for $0 < \varepsilon \leq \Delta(0)$,

$$R_\varepsilon^* = \frac{1}{2} \log_2 \frac{\Delta(0)}{\varepsilon}.$$

If $\Delta(0) = 0$, no communication is needed.

Proof. Center variables without loss of generality. The coalesced conditional mean is $T' = \mathbb{E}[X | Y^A, Y^B] = a^\top Y^A + b^\top Y^B$. Let $V = b^\top Y^B$, computable by the encoder. Orthogonality of the conditional-mean residual gives, per sample and hence averaged over a block,

$$\mathbb{E}(X - \hat{a})^2 = \mathcal{L}_{\text{coal}} + \mathbb{E}(T' - \hat{a})^2.$$

Because the decoder knows $a^\top Y^A$, the excess term is the distortion in estimating scalar V with decoder side information Y^A . Gaussian Wyner–Ziv coding has no rate loss relative to conditional rate–distortion under quadratic loss [4, 5], giving $D(\kappa) = \text{Var}(V | Y^A) 2^{-2\kappa}$. For the converse, giving Y^A to the encoder cannot increase the minimum distortion; conditional Gaussian rate–distortion already gives this same lower bound, including encoders that retain all of Y^B . Finally, the orthogonal decomposition

$$X - \mathbb{E}[X | Y^A] = (X - T') + (V - \mathbb{E}[V | Y^A])$$

shows $\text{Var}(V | Y^A) = \Delta(0)$. Substitution and inversion give the claims. $\square$

Corollary 1 (Exact information bridge in the benchmark). *For positive coalesced residual variance in Theorem 1,*

$$I(X; Y^B | Y^A) = \frac{1}{2} \log_2 \frac{\text{Var}(X | Y^A)}{\text{Var}(X | Y^A, Y^B)}, \quad \Delta(0) = \text{Var}(X | Y^A) (1 - 2^{-2I(X; Y^B | Y^A)}).$$

This is an application of classical Gaussian source coding, not a new source-coding theorem. The fourfold reduction per bit is exact only in this benchmark. One-shot handoffs, discrete actions, non-Gaussian tasks, finite blocks, restricted policies, and suboptimal codecs can have different curves. Block-coding latency is also an economic cost, even though it is absent from the rate–distortion expression. Capacity requirements grow logarithmically with tolerance in this model, but acquiring capacity need not be cheap.

7.3 The economic switching threshold

For one fixed cut, let all non-decision costs use the same objective units as expected loss. Let C_{coal} and C_{split} include communication, computation, interpretation, delay, governance, and any amortized transition costs. Set

$$H = C_{\text{coal}} - C_{\text{split}}.$$

For this comparison surprise is diagnostic ($\beta_S = 0$), or any separately justified predictive burden is already included in the stated costs. Then

$$J_{\text{split}} - J_{\text{coal}} = \Delta(\kappa) - H.$$

Corollary 2 (Economic switching threshold). *Under Theorem 1, suppose $\Delta(0) > 0$ and coordination savings $H > 0$ are held fixed as κ varies. Splitting is weakly preferable exactly when*

$$\kappa \geq \kappa_{\text{economic}} := \max \left\{ 0, \frac{1}{2} \log_2 \frac{\Delta(0)}{H} \right\}.$$

At a positive threshold the architectures tie; above it splitting is strictly preferable. If $H > \Delta(0)$, splitting is strictly preferable even at zero capacity; if $H = \Delta(0)$, zero capacity is a tie.

Proof. Substitute the deficit benchmark into $J_{\text{split}} - J_{\text{coal}} \leq 0$ and solve $\Delta(0)2^{-2\kappa} \leq H$ for $\kappa \geq 0$. $\square$

If $H < 0$, splitting cannot win under this fixed-cut comparison. If $H = 0$ and $\Delta(0) > 0$, no finite rate attains a tie in the Gaussian benchmark. When $\Delta(0) = 0$, compare the costs alone. More generally, capacity costs or coding delay make $H = H(\kappa)$; then optimize or solve $\Delta(\kappa) \leq H(\kappa)$ rather than apply the closed form blindly.

For example, with $\Delta(0) = 4$ loss units and $H = 1$ cost unit per decision, the threshold is one bit. At that rate the remaining decision deficit is one unit: the interface is lossy, but its savings exactly compensate. At two bits, the deficit is 1/4 and splitting improves the objective by 3/4 unit.

An information-clean boundary loses no decision value. An economically efficient boundary loses no more decision value than its coordination savings justify.

7.4 Granularity in a balanced-unit cost model

A separate model makes the number of units explicit. Let N primitive components be divided into m equal-sized units. Assume within-unit coordination cost an^2 for size n , $a > 0$, and a sparse, tree-like quotient graph with $m - 1$ interfaces costing $B > 0$ each. The continuous balanced-size relaxation is

$$\Theta(m) = \frac{A}{m} + B(m - 1), \quad A = aN^2, \quad 1 \leq m \leq N.$$

Quadratic internal cost is a modeling assumption motivated by possible all-pairs interactions, not a claim that all coordination is all-pairs [13, 8]. Balanced sizes, sparse interfaces, and constant per-interface costs are separate assumptions. Information-clean cuts do not by themselves imply any of them. Decision loss must be constant or separately modeled for minimizing Θ to minimize the full objective.

Theorem 2 (Optimal decomposition in the continuous relaxation). *For the stated cost model,*

$$m_{\text{cont}}^* = \min\{N, \max\{1, \sqrt{A/B}\}\}.$$

At an interior optimum, $\Theta(m^) = 2\sqrt{AB} - B$. For integer m in the same surrogate cost, evaluate the feasible floor and ceiling of the clipped continuous optimum and choose the lower-cost one.*

Proof. $\Theta'(m) = -A/m^2 + B$ and $\Theta''(m) = 2A/m^3 > 0$. Clipping the stationary point gives the constrained solution; convexity leaves only its adjacent feasible integers to compare. $\square$

For integer steps in this surrogate, introducing one more unit reduces cost exactly when

$$\Theta(m + 1) - \Theta(m) = B - \frac{A}{m(m + 1)} < 0.$$

When indivisible primitives prevent equal sizes, the exact internal cost for balanced integer groups is $a \sum_j n_j^2$, with group sizes differing by at most one. Optimize that discrete expression rather than claiming the continuous surrogate is exact. The square-root form is the familiar opposing-cost structure, not a universal organizational scaling law.

Corollary 3 (Relative cost changes shift granularity). *Within the interior regime of the same cost model,*

$$\frac{m^{*'}}{m^*} = \sqrt{\frac{A'/A}{B'/B}}, \quad \Delta \log m^* = \frac{1}{2}(\Delta \log A - \Delta \log B).$$

Holding A fixed, halving B increases the continuous optimal count by $\sqrt{2}$, approximately 41%. If both costs change, finer decomposition is favored when interface costs fall proportionally more than internal costs; the reverse can favor integration. Clipping and integer constraints can leave the implemented count unchanged.

Internal coordination burden A is distinct from cross-boundary decision dependence. Raising A alone favors splitting in this model; raising the loss caused by a cut can favor coalescence. A single “coupling” slider cannot be interpreted as both without an explicit mapping.

Remark 1 (Lossy interfaces and interacting deficits). *With lossy cuts use a total excess decision-loss term $D(\mathcal{B}, \kappa)$:*

$$J(\mathcal{B}, \kappa) = \mathcal{L}_{\text{coal}} + C(\mathcal{B}, \kappa) + D(\mathcal{B}, \kappa).$$

Single-cut deficits generally cannot be added without assumptions excluding shared, redundant, or interacting information. In a specified nested family with surrogate D_m , adding a unit is worthwhile only if $\Theta(m+1) - \Theta(m) + D_{m+1} - D_m < 0$. Increasing these marginal deficits opposes further splitting; a positive deficit by itself does not prohibit it. A tolerance constraint $D_m \leq \varepsilon$ is a separate governance choice, not an economic necessity.

For a general within-unit cost f , use $mf(N/m) + B(m-1)$. Strict convexity of f does not guarantee an interior optimum or a power-law solution. For the particular family $f(n) = an^p$, $p > 1$, the interior solution is $m^* = [a(p-1)N^p/B]^{1/p}$. A dense-interface approximation $A/m + Bm^2$ instead yields $(A/(2B))^{1/3}$. These examples make the assumptions and topology dependence explicit [24, 26]. Sequential organization design additionally requires transition, learning, and delay costs; the static results alone do not solve that problem.

8 Organizational Agility

Speed alone is not agility. An organization can act quickly and incorrectly. Correctness alone is not agility either. An organization can make the right decision too late.

Let $\tau_t > 0$ be the end-to-end latency from signal availability to coordinated action, measured in a fixed time unit. Let a_t^0 be a baseline action, default policy, or status-quo response. Define decision value as loss reduction relative to that baseline:

$$Q_t = \mathbb{E} [L(X_t, a_t^0) - L(X_t, a_t)].$$

The quantity Q_t may be negative when the coordinated action performs worse than the baseline; in that case, speed amplifies negative decision value rather than producing agility. Organizational agility is decision value per unit time:

$$\mathcal{G}_t = \frac{Q_t}{\tau_t}.$$

A cost-adjusted version is

$$\mathcal{G}_t^{(c)} = \frac{Q_t}{\tau_t(1 + \tilde{C}_t)},$$

where $\tilde{C}_t = C_t/C_0 \geq 0$ uses a declared, fixed resource-cost scale $C_0 > 0$. Comparisons use the same baseline policy, loss, time unit, and cost scale. This ratio is a diagnostic, not generally an objective equivalent to minimizing J . For non-overlapping sequential episodes, a value-rate aggregate is $\sum_t Q_t / \sum_t \tau_t$, not an unweighted average of episode ratios. Concurrent workflows require an explicit throughput and resource-accounting convention.

This definition makes the quality term explicit. An action can encode a great deal of information about the state and still be a poor action. Loss reduction, expected utility improvement, or expert-rated decision value is therefore the primary quantity. Mutual information remains useful as an idealized proxy:

$$I(X_t; a_t)$$

is a measure of information fidelity between the world and the organization’s actions. It is appropriate when higher information fidelity monotonically improves action quality under the policy class being studied. Otherwise, decision value should dominate.

9 Compression and Organizational Structure

Organizations rarely transmit the full state X_t . Instead, they construct internal representations

$$X_t \rightarrow Z_t \rightarrow a_t.$$

Roles, dashboards, metrics, status reports, tickets, runbooks, APIs, schemas, plans, OKRs, approvals, and incident summaries are compression mechanisms. They reduce communication burden, but they can also destroy decision-relevant information. A structured incident ticket, for example, is a protocol that compresses incident state across a boundary: it can preserve severity, affected service, suspected cause, owner, and customer impact, or it can omit the diagnostic detail needed for a good response.

The organizational design problem has the structure of a rate-distortion problem:

$$\min_f C_{\text{repr}}(f) + \lambda \mathbb{E}[L(X_t, a_t(f(X_t)))],$$

where $C_{\text{repr}}(f)$ is the cost of maintaining and communicating representation f , and L is decision loss. A dashboard that compresses a complex operational state into one green/yellow/red indicator is valuable if it preserves the information needed for action. It is dangerous if it hides the structure needed for diagnosis.

10 Software Agents as Boundary-Shifting Infrastructure

Software agents are not merely labor-saving tools. They are boundary-shifting infrastructure.

A software agent can be modeled as a node

$$\text{Agent}_i = (\mathcal{O}_i, \mathcal{M}_i, \pi_i, \mathcal{T}_i, \Gamma_i),$$

where $\mathcal{O}_i$ is the set of observation channels, $\mathcal{M}_i$ is the message schema or protocol, π_i is the local decision policy, $\mathcal{T}_i$ is the set of tools or actuators, and Γ_i is the governance envelope: permissions, constraints, auditability, and escalation rules.

The important organizational effect of software agents is not only task automation. It is that they make parts of cross-boundary coordination programmable. They can observe, compress, route, escalate, execute, and audit across interfaces that previously required human interpretation or managerial hierarchy. When sensing, routing, interpretation, escalation, and execution can be mediated by machine-readable protocols, the efficient boundary between hierarchy, market, and hybrid organization changes.

Agents can observe continuously, summarize and route information, enforce schemas, call tools, monitor queues, escalate uncertainty, and execute routine actions. But agentic coordination is not automatically superior. It may increase local-global misalignment, proxy optimization, monitoring burden, structured propagation of bad assumptions, and compression loss from over-summary.

The relevant comparison is conditional:

Agents improve organizational agility when tasks are sufficiently structured, protocols are sufficiently expressive, and governance is strong enough to control misalignment.

In an agent-mediated incident workflow, a software agent can summarize evidence, route exceptions, trigger routine remediation, and escalate uncertainty without requiring every actor to merge into one permanent team.

11 Running Example: Incident Response

Incident response is a canonical instance of the agent-boundary problem. The latent state X_t is the true incident state: affected service, root cause, blast radius, customer impact, and time sensitivity. Local observations Y_t^i include alerts, logs, metrics, customer reports, traces, and partial diagnoses held by SRE, support, infrastructure, product, vendor, or agentic monitoring nodes. Local representations Z_t^i include each actor’s working diagnosis. Messages M_t^{ij} include pages, Slack messages, ticket updates, dashboard links, incident summaries, and agent-generated briefings. Actions a_t include rollback, mitigation, escalation, customer communication, vendor failover, or no-op.

The cost decomposition becomes concrete, as summarized in Table 2.

Table 2: Incident response as an agent-boundary problem

Formal term	Incident-response interpretation
X_t	True incident state: root cause, severity, affected users, blast radius.
Y_t^i	Logs, metrics, alerts, customer reports, traces, vendor status.
Z_t^i	Local diagnosis held by SRE, support, infra, product, vendor, or agent.
M_t^{ij}	Pages, Slack updates, tickets, dashboards, agent summaries, API events.
a_t	Rollback, mitigation, escalation, customer communication, failover.
C_{interp}	Translation cost across SRE, support, product, vendor, and customer-facing language.
C_{delay}	Outage duration, customer impact, missed SLA, lost trust.
C_{error}	Wrong diagnosis, bad rollback, repeated mitigation, incident reopen.
C_{misalign}	Support says one thing while infrastructure does another.
C_{monitor}	Approval gates, blast-radius review, compliance and audit checks.

The boundary choices are visible. One integrated incident room coalesces actors into a shared representation and control loop. Separate teams preserve local autonomy but communicate through

tickets and status updates. A vendor boundary uses contracts, SLAs, and escalation paths. An agent-mediated hybrid uses software agents to monitor queues, summarize evidence, enforce schemas, route exceptions, and trigger routine tool calls while escalating uncertainty to humans.

The best boundary depends on interdependence, protocol quality, volatility, observability, and governance. If the incident is tightly coupled across systems and the interfaces are weak, coalescence is attractive. If the incident is modular and protocols are strong, split teams or vendors can coordinate effectively. If the workflow is semi-structured, observable, and governed, software agents can absorb routing and interpretation burden without requiring full organizational merger.

12 Design Implications and Conditional Hypotheses

Proposition 2 (Protocol emulation). *At fixed architecture, suppose every policy feasible under protocol P can be emulated under P' with the same conditional action distribution and no greater total non-decision cost. Then the minimum net objective under P' is no worse than under P .*

Proof. Emulate an optimal old policy (or an arbitrarily close approximation); optimization over the new feasible set cannot do worse. $\square$

A Blackwell improvement in information can supply the decision component of this condition when the policy class is closed under the required garbling [3]. It does not establish the cost component: a richer schema can add interpretation, computation, review, or delay. Informativeness alone is therefore insufficient for a net-objective claim.

Proposition 3 (Latency under volatility). *Suppose a relevant state survives a delay τ with probability $e^{-\lambda\tau}$, useful decision value is zero after a state change, and $Q_0 > 0$ is held fixed. Then $Q(\tau) = Q_0 e^{-\lambda\tau}$. Greater hazard λ increases the proportional sensitivity to delay: $-\partial \log Q / \partial \tau = \lambda$. The absolute marginal benefit of reducing delay is $Q_0 \lambda e^{-\lambda\tau}$ and increases with λ only while $\lambda\tau < 1$.*

Proof. Condition on state survival and differentiate:

$$\frac{\partial}{\partial \lambda} \left(-\frac{\partial Q}{\partial \tau} \right) = Q_0 e^{-\lambda\tau} (1 - \lambda\tau).$$

Once information is already very stale, a small latency improvement has little absolute value. The exponential form assumes a constant survival hazard; it is not a general theorem about all dynamic decision problems. $\square$

Interdependence and modularity: a conditional hypothesis. Greater cross-boundary decision dependence can raise excess loss and favor coalescence, whereas lower interface costs can favor separation. This is not a universal monotone relation with an undifferentiated “coupling” variable. Internal coordination burden A and cross-boundary decision deficit D play different roles: increasing A alone favors splitting in the balanced-unit model, while increasing the marginal deficit of additional cuts opposes it. The economic comparison is Corollary 2, not information cleanliness alone.

Remark 2 (Agentic routing: net-objective accounting). *Let ΔL and ΔC_j denote agent-assisted minus baseline expected loss and each disjoint cost component, respectively, in the same evaluation horizon. With fixed objective weights and $\beta_S = 0$, the exact net-objective condition is*

$$\Delta J = \Delta L + \sum_j \lambda_j \Delta C_j < 0.$$

Any independently priced predictive burden adds $\beta_S \Delta S$. Execution effort belongs in computation cost; measurable reviews and approvals belong in monitoring cost. Broader governance burdens must not be counted again under overlapping labels. Routing or execution savings alone are insufficient if direct decision loss or other costs increase by more.

An improvement in this objective does not automatically improve the agility ratio. That comparison separately requires $Q'/[\tau'(1 + \tilde{C}')] > Q/[\tau(1 + \tilde{C})]$, with a common baseline, time unit, and cost scale. Neither information fidelity nor faster execution alone is sufficient.

Governance limit: a conditional hypothesis. Agentic decentralization can have diminishing or negative returns when monitoring, unsafe actions, or local–global misalignment grow faster than coordination savings. A fast local optimizer is useful only insofar as its actions improve the declared global objective. The framework identifies terms to measure; their signs and magnitudes require evidence.

13 Executable Illustration

The companion is an executable illustration, not an empirical validation or a calibrated causal estimate. Its original Monte Carlo scenario model uses chosen noise and cost coefficients to compare coalesced, split, market, and hybrid architectures. Those scenarios do not implement optimal Wyner–Ziv codes, infer operational bit rates from schema sliders, or optimize arbitrary partitions. Their phase map is a sensitivity visualization for that heuristic model, not an estimated information-theoretic boundary.

A separate analytical workbench evaluates $\Delta(\kappa) = \Delta(0)2^{-2\kappa}$, the economic switching condition, and the balanced-unit cost curve over integer unit counts. It makes the assumptions visible and separates a numerical evaluation of a formula from experimental evidence. Decision-value agility in the scenario model uses the zero-action baseline under squared loss and retains negative values; mutual information and sign accuracy are diagnostics, not the quality numerator. The original scenario architecture still has a fixed three-group split; the integer granularity sweep belongs to the separate workbench.¹

Figure 1 depicts the economic comparison directly, in common objective units. It is an analytical decision diagram, not a simulation result.

The heuristic simulation also exposes a planning tradeoff: up-front representation investment can lower runtime error while adding delay or rigidity. The chosen coefficients illustrate that possibility; they do not establish its empirical magnitude or direction in a particular organization.

14 Measurement and Empirical Agenda

The empirical goal is to determine whether a boundary or protocol change improves a specified decision objective, and whether the measured improvement comes from better information, lower resource cost, lower delay, or a combination. No empirical validation is claimed in this draft.

14.1 A controlled replay as the first test

A tractable first study uses incident or support cases with a defensible reference outcome and the evidence available at the original decision time. Freeze the underlying cases, loss rubric, downstream

¹Companion: galatheus-labs.github.io/coase-info-theory. Analytical workbench: analytical-workbench.html in the [public repository](#). Draft-branch changes are not live until merged and deployed.

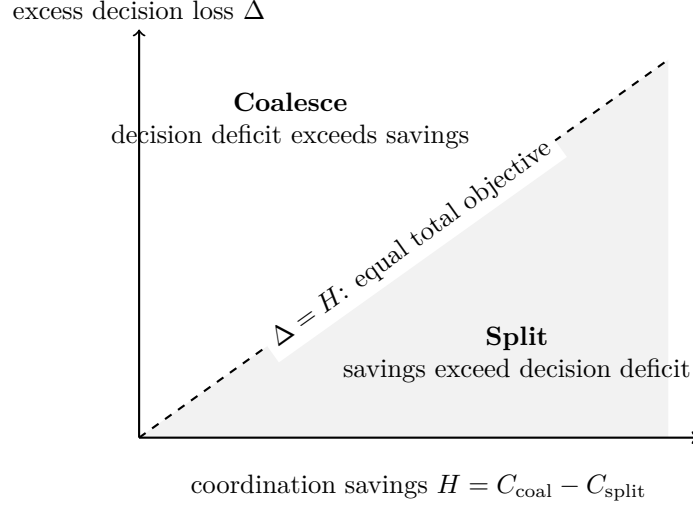


Figure 1: Economic, not lossless, switching. Both axes use the same objective units per decision; the drawing uses different display scales on the two axes. For $H > 0$, splitting is weakly preferable below $\Delta = H$, even with positive excess loss. In the Gaussian benchmark at fixed H , increasing capacity lowers Δ and can cross this line at the finite threshold in Corollary 2.

policy or model version, permissions, tool access, and evaluation horizon. Compare four information conditions on the same cases: local evidence only; a fixed structured handoff; an agent-mediated handoff under a declared communication budget; and a full-evidence reference. The full-evidence reference is a practical comparator, not automatically the Bayes-optimal coalesced risk.

Vary the permitted handoff representation or budget independently of case complexity. Pre-specify randomization, paired evaluation, stochastic repeats, and blinded outcome scoring where feasible. Agent mediation should not gain extra tools or action authority unless those are separate experimental factors. Prevent hindsight leakage: later postmortems can label an outcome but should not become evidence available to a policy at the earlier decision point.

Report direct decision loss, end-to-end latency, compute and token expenditure, human interpretation effort, review effort, and safety outcomes separately before combining them with declared objective weights. Include reference-policy loss on every case, uncertainty intervals on paired differences, and sensitivity to the loss rubric and cost weights. An observed reference gap can be negative if the full-evidence policy is not optimal; do not clip such gaps or silently take their logarithms. Choose sample size from a pilot variance, clustering structure, and a pre-specified effect of interest, not a calendar quarter alone. Cluster resampling by incident or another defensible independent unit when multiple observations share a case.

This tests a comparative boundary-design hypothesis, not the Gaussian coding law by default. A separate synthetic Gaussian experiment can verify the analytic expressions or compare finite-budget codecs with the asymptotic benchmark. It must not label a suboptimal codec’s finite-sample curve an exact realization of that bound.

14.2 Operational proxies and their limits

Clarification round-trips and schema completeness can diagnose interface insufficiency, but they do not estimate R^* and κ in common units without a separate validated measurement model. Do not subtract them to construct $R^* - \kappa$. Clarification is partly an outcome of the interface; treating it as

Table 3: Measurements, proxies, and interpretation limits

Quantity	Operational treatment
Decision value Q	Paired loss reduction against a fixed baseline; expert scores, avoided rework, or SLA outcomes need a declared loss mapping.
Predictive surprise S	Proper log scores on fixed targets where predictive distributions and labels exist; escalation and reopen events are diagnostics, not themselves Shannon surprise.
Latency τ	Signal-to-triage, signal-to-decision, and signal-to-action intervals reported separately in a fixed unit.
Interpretation cost	Clarification requests and active translation time; request count is affected by task difficulty and protocol inadequacy.
Interface richness	Field availability, missingness, representation type, and actual message budget; completeness fractions are not bit capacities.
Monitoring cost	Review minutes, approvals, audits, and escalations, with non-overlapping accounting.

an exogenous communication requirement can make the proposed test circular.

Operational bits require a declared alphabet, code, and per-decision or per-block rate convention. A token count is an engineering budget, not a semantic information rate by definition. Field counts and completeness fractions cannot test a fourfold-per-bit slope. Even a defensible bit measure does not make reopen probability or resolution time a Gaussian quadratic loss. Theorem 1 supplies a benchmark; transfer of its functional form to workflow outcomes is an additional hypothesis.

Likewise, Proposition 1 does not predict an empirical elbow. $R_\varepsilon^\star$ is an operational tolerance crossing that can occur on a smooth curve. Compare flexible alternatives before claiming a threshold or log-linear relationship, and estimate an excess-loss curve only against a well-defined reference. The economic comparison is whether the loss increase is outweighed by measured coordination savings in common units.

For continuity with the trace playbook, a descriptive proxy can be written

$$\hat{G} = \frac{Q}{t_0(1 + T_{\text{resolution}}/t_0)(1 + \alpha H_{\text{handoff}})}, \quad t_0 > 0, \quad \alpha \geq 0.$$

Here Q is baseline-relative value, t_0 is a fixed reference time, and H_{handoff} is a handoff count, distinct from the economic savings H . Declare t_0 and α , retain negative values, and report the components separately. This stabilized proxy is neither a literal information-rate estimate nor generally equivalent to minimizing J . Avoid counting handoff penalties again when their full costs already enter the objective.

14.3 Observational deployment studies

A matched before/after analysis can characterize associations after adjusting for measured case mix. It does not eliminate unobserved selection, changing staffing, incident severity, temporal trends, or simultaneous tooling changes. Staggered adoption can support a stronger design only with stated identification assumptions, an appropriate estimator, checks on pre-treatment behavior, and attention to spillovers between teams. Randomized rollout or crossover, where safe, can more

directly isolate the interface intervention. Safety and permission constraints remain part of the feasible experimental design.

To examine the granularity model, compare costs across alternative feasible partitions, not merely the number of teams an organization happens to have. Estimate changes in both A and B , record cross-boundary deficits and transition costs, and check the balanced-unit and sparse-interface assumptions. The conditional prediction is $\Delta \log m^* = \frac{1}{2}(\Delta \log A - \Delta \log B)$ within the interior surrogate regime. Adoption alone does not predict a 41% increase or finer decomposition. Register the planned analysis before examining test outcomes; this proposed protocol has not itself been externally preregistered.

15 Limitations

This framework has several limitations.

First, decision value is difficult to measure. Expert labels, SLA outcomes, severity reduction, and rework are proxies, not direct access to $L(X_t, a_t)$.

Second, mutual information is elegant but subordinate. It is useful for idealized information-fidelity arguments, but high mutual information between state and action does not guarantee good action.

Third, agents do not universally improve coordination. They help most when workflows are structured, observability is high, and governance is clear.

Fourth, some organizational knowledge is tacit and difficult to encode into protocols. Over-protocolization can compress away context.

Fifth, the current simulation is a stylized model. It illustrates mechanisms but does not yet estimate causal effects.

Sixth, organizational utility is multi-objective. Speed, accuracy, compliance, safety, morale, trust, and strategic learning may trade off.

Seventh, the Gaussian benchmark requires scalar quadratic estimation, memoryless samples, optimal coding, and asymptotically long blocks. A real-time agent handoff need not approach that bound; block delay must be costed.

Eighth, balanced units, sparse interfaces, and constant interface costs are separate assumptions. Heterogeneous tasks and interacting communication cuts can invalidate the square-root model and additive deficit approximations.

Finally, this is a normative design framework, not a theory of equilibrium ownership or bargaining. Strategic behavior, transition costs, organizational learning, and distributional consequences require explicit extensions.

16 Conclusion

Coase–Information Theory reframes the firm as an information architecture and then generalizes that idea. The key economic question is not only whether work should be coordinated by markets or hierarchy, but which agents should coalesce, which should split, and which should coordinate through protocols so that distributed observations become coordinated action.

Operationally, the framework suggests asking four questions before changing a boundary. What decision-relevant information is lost at the interface? What delay is introduced by the current boundary? What monitoring or misalignment cost would a new boundary create? Can a software agent or protocol preserve enough state to avoid full organizational coalescence?

Organizational agility is the rate at which decision-relevant information becomes valuable action. Software agents matter because they can change the latency, fidelity, and cost of coordination. Their importance is not simply that they automate tasks. It is that they alter where boundaries should be drawn: inside firms, across teams, between vendors, and between human and machine actors. They are boundary-shifting infrastructure.

The central economic distinction is between preserving all decision value and preserving enough to justify a boundary's cost. A lossy interface can be optimal. Agents can move the optimum in either direction by changing internal coordination, interface costs, and the value of shared context.

The next generation of AI-native companies may be designed not merely around people using AI tools, but around software-mediated sensing, routing, decision, and execution loops. The purpose of Coase-Information Theory is to provide a language for designing and measuring those loops.

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